

Preliminary Evaluation of Piezoelectric Thin Film Sensors for Arterial and Venous Pulse Detection Using an Epidermal Neck Patch

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Abstract

Wearable technology for monitoring blood pulse waveforms in the cervical area is still in a developmental stage, primarily due to challenges in accurate sensor placement and proper signal detection on the neck. In this study, two commercial piezoelectric thin film sensors were compared to identify the most suitable one for integration into a biocompatible epidermal neck patch. Of the two, the one with the largest sensitive area was chosen to carry out measurements on a healthy male subject in two different body positions: supine and sitting. The ECG signal was also acquired as a temporal reference to distinguish physiological blood pulse waveforms from those containing artifacts. Signal processing techniques applied to recorded signals have demonstrated the ability of the piezoelectric thin film sensor to clearly distinguish between arterial and venous blood pulses. Moreover, thanks to the used non-disposable adhesive patch, these preliminary results pave the way for telemedicine applications focused on head and neck vascular health, enabling the screening of population and contributing to personalized healthcare and overall public wellbeing.

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CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**

Keywords

Carotid Arterial Pulse, Jugular Venous Pulse, Piezoelectric Thin Film Sensors, Epidermal Patch, Pulse Wave Analysis

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1 Introduction

The cervical area plays a key role in cardiovascular health due to the presence of major vascular structures, such as carotid arteries and internal jugular veins. Monitoring the blood pulse of these vessels is essential for screening and prevention of cardiovascular disease, which remain the leading cause of death worldwide according to data from the World Health Organization (WHO) [1].

Common carotid arteries (CCA) are essential for cerebral perfusion, and their involvement in atherosclerotic processes significantly increases the risk of ischemic stroke [2]. Similarly, internal jugular veins (IJV) are key conduits for cerebral venous drainage, and their dysfunction can contribute to altered intracranial hemodynamics [3]. Vascular pathologies in the neck and head are not only markers of systemic cardiovascular disease but also potential contributors to cerebrovascular events [4,5].

In this context, we want to focus on monitoring vascular parameters in the cervical area. Particularly, blood pressure can be measured at the level of the common carotid artery while the jugular venous pressure (JVP) that is defined as the estimation of the right atrial pressure, can be directly observed through the IJV [6-8]. Several techniques are used to assess these parameters, such as sphygmomanometer, ultrasound, MRI, CT scan, and, slightly more invasively, catheter to detect venous pressure [9]. However, measuring cardiovascular parameters on the neck is not usual without the help of a healthcare operator, so in most cases the patient must go to the clinics to perform the medical examination.

Modern trends in wearable technology focus on non-invasive and continuous health monitoring, including bio-signals on areas of the body that are difficult to access due to the irregular shape of the skin [10,11]. Wearable epidermal sensors, also known as electronic patches, enable non-invasive and unobtrusive health monitoring. These devices are designed to conform closely to the skin, allowing the detection of physiological waveforms with minimal discomfort [12]. Their success largely depends on the materials used for their development, ensuring flexibility, biocompatibility, and long-term adhesion.

In this work we want to test the possibility to monitor cardiovascular parameters on the neck with an epidermal patch, which consists of a commercial piezoelectric thin film sensor and an adhesive silicone gel as interface between the sensor and skin. Although several studies in the literature propose epidermal patch prototypes for carotid artery blood pressure monitoring, no work has yet fully demonstrated the ability of this technology to detect the jugular venous pulse (JVP), which provides crucial insights into both cardiac activity and potential neurodegenerative disorders.

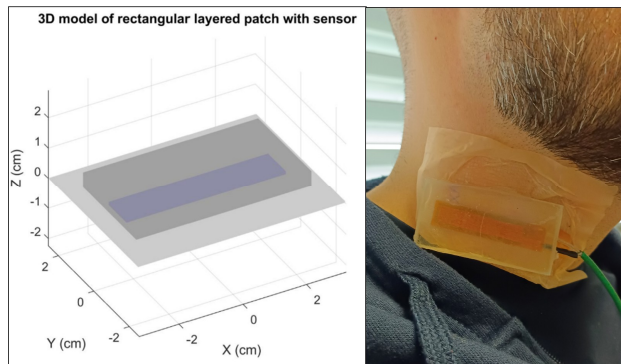


Figure 1: Epidermal neck patch. On the left side is the 3D model design of the patch. On the right side is the patch on the J2 segment of the right IJV.

2 Materials and Methods

To provide proof of concept for the use of a piezoelectric thin film sensor in assessing cardiovascular parameters on the neck surface, we compared two sensors made by the same piezoelectric PIC 255 material, but with different geometries. They are from the DuraAct Patch Transducers family, PI Ceramic GmbH [13]. The two models selected for this study were the P-876.SP1 with a sensitive

area of 10 mm × 10 mm size and 200 μm thickness, and the P-876K015 with sizes of 40 mm × 09 mm × 200 μm.

To fabricate the epidermal patch, each piezoelectric sensor was sandwiched between two layers of biocompatible adhesive silicone gel (Zhermack SpA). The bottom layer serves as the sensor-to-skin interface, while the upper layer secures the sensor in place.

Then, the two piezoelectric sensors were placed on different neck positions to compare the signal output between them. Particularly, as we are mostly interested in demonstrating the ability of the piezoelectric thin film sensor to detect the JVP waveform, the point of application of the sensors for the signal comparison where J1 and J2 segments of right IJV, and the sternum level.

Once the best piezoelectric thin film sensor and the most suitable neck location had been selected for the experimental phase (Figure 1), blood pulse measurements were taken from a healthy subject in two different positions: supine and sitting. This is due to the different body positions causing variations in blood flow within the jugular veins, as their walls collapse under the effect of gravity. Indeed, it is well known that blood flow in the jugular veins is greatest when a person is in supine position [14].

The measurements were acquired with the handheld RTH1004 Digital Oscilloscope, up to 5 GSa/s of sampling rate, Rohde & Schwarz.

During the experiment, it was essential to synchronize the electrocardiogram (ECG) signal with the signal recorded by the piezoelectric thin film sensor placed on the subject's neck [15,16]. A three-lead ECG system [7] was used to detect the changes in heart rhythm. To ensure accurate synchronization during data collection, a physical artifact in the form of a simultaneous finger tap on both systems was introduced. Therefore, it was possible to align the data in post-processing.

The acquired data were analyzed by using MATLAB R2024b software, MathWorks. After the synchronization between ECG and piezoelectric signals, the data were pre-processed in the form band-pass filtering, normalization, and time windows selection to isolate representative signal segments. Next, pulse wave analysis was performed separately on the signals acquired with the subject in the supine position and those acquired while sitting.

For the subject in the sitting position, the acquired signals required a pulse wave analysis approach like those used in processing photoplethysmography (PPG) data [17,18], which are associated with arterial blood flow. Indeed, physiological landmarks such as the systolic peak and wave onset were extracted from the filtered signal. To further analyze pulse morphology and compute vascular health metrics, the filtered signals were subjected to first, second, and third-order numerical derivatives [17]. This enabled the calculation of physiological metrics such as ratios between fiducial points, slope delays, and the vascular aging index (VAI) for the subject involved in the test.

Conversely, the waveform of the signal detected when the subject was in the supine position is expected to reflect blood flow in the internal jugular vein (IJV), also known as the jugular venous pulse (JVP). This is a venous pulse, which is fundamentally different from the arterial one. For this reason, it is important to use the ECG

signal as a temporal reference to properly identify the five characteristic peaks of the JVP [19,20], as they are a consistent feature of every heartbeat. Currently, there is no established signal processing technique used to analyze the JVP waveform. Thus, pulse wave analysis for these signals aims to expand existing information about cardiac events within a heartbeat, by calculating parameters based on time differences between the peaks of the JVP and the peaks of the ECG. Thus, the piezoelectric filtered signals were subjected to mathematical integral, according to the methodology proposed in [21], then the instant of time for both the JVP peaks and the ECG peaks were used to compute the time differences.

3 Results

While choosing the best piezoelectric thin film sensor between the P-876.SP1 and the P-876K015, we figured out that even a small change in the sensor location has a large effect on the signal obtained for the P-876.SP1 compared to the P-876K015. Indeed, since the P-876K015 has a larger sensitive area, it was more successful because it is not as susceptible to being precisely placed, whereas the P-876.SP1 needs to be very accurately placed on the neck. While comparing the different points of application of the piezoelectric sensors on the neck, we chose the J2 segment as the measured voltage output was the largest compared to that from the J1 segment and the sternum.

Figure 2 shows the signal outputs acquired with both sensors placed at the J2 segment of the right internal jugular vein. It is clearly visible that the signal output from the P-876K015 has a larger peak-to-peak amplitude than that of the P-876.SP1. The waveforms in Figure 2 are plotted with a horizontal offset, where the x-axis time values are displayed with a minus sign.

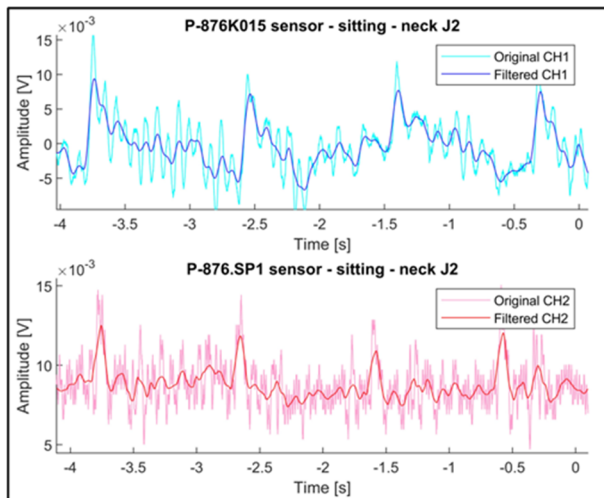


Figure 2: Signal output comparison between P-876K015 and P-876.SP1 piezoelectric thin film sensor on J2. Both signals are plotted with a horizontal offset, and they were filtered using the Gaussian filter on MATLAB R2024b.

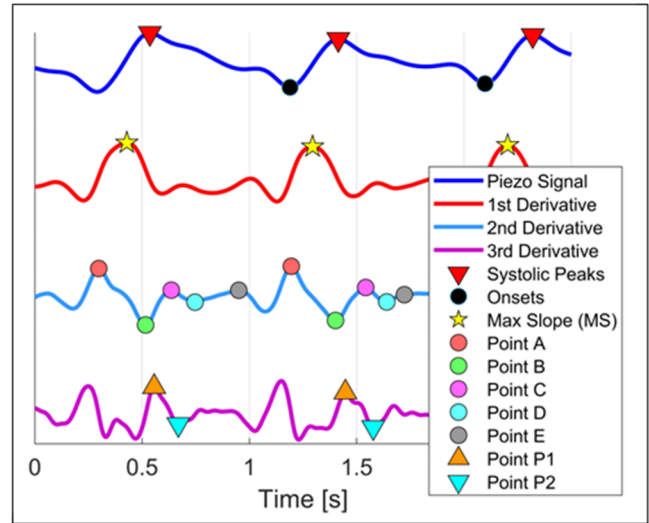


Figure 3: Graphical result of the data analysis carried out on the signal measured when the subject was seated. The waveforms are plotted in arbitrary units.

The experiment was conducted using only the P-876K015 sensor on a healthy 37-year-old male, BMI: 24.2. The measurements were repeated twice for both the body positions chosen for the test: sitting and supine. Each measurement lasts for approximately thirty seconds. The heartbeat of the subject was 72 bpm.

Figure 3 shows the graphical results obtained from the data analysis of the signal acquired with the subject in the sitting position, while Figure 4 corresponds to the signal acquired with the subject in the supine position. In both figures, only a few cardiac cycles are plotted to better show the waveform of the signals and to highlight the detected point of interest. For Figures 3 and 4, the y-axis is expressed in arbitrary units.

The fiducial points identified on the waveforms plotted in Figures 3 and 4 were used to calculate the cardiovascular parameter values listed in Table 1 and Table 2, respectively.

Table 1: Results of data analysis for measurements with the subject in sitting position – Carotid Arterial Pulse Waveform.

Parameter	Values, P-876K015	Values, literature
b/a	-0.63 ± 0.09	-0.81 ± 0.11 [22]
c/a	0.19 ± 0.05	0.01 ± 0.12 [22]
d/a	-0.07 ± 0.03	-0.17 ± 0.15 [22]
e/a	0.13 ± 0.01	0.21 ± 0.09 [22]
MS Delay (s)	0.10 ± 0.01	0.08 ± 0.14 [23]
P2-P1 Delay (s)	0.13 ± 0.02	0.11 ± 0.15 [24]
P2/P1 Ratio	0.39 ± 0.06	0.2 ± 0.4 [24]
VAI	-0.88 ± 0.16	-0.85 ± 0.12 [22]

Tables 1 and 2 compare the obtained results with values reported in the literature, confirming the effectiveness of the chosen piezoelectric thin film sensor as a sensitive element for developing

an electronic system in the form of an epidermal patch. In Table 2, the results are expressed as a fraction of the cardiac cycle (ccf).

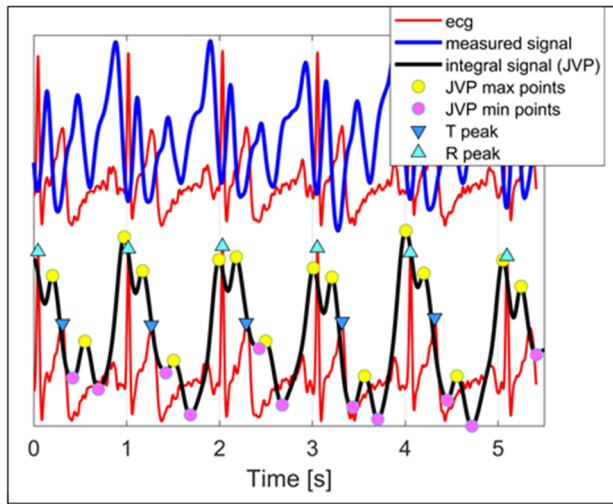


Figure 4: Graphical result of the data analysis carried out on the signal measured when the subject was supine. The waveforms are plotted in arbitrary units.

Table 2: Results of data analysis for measurements with the subject in supine position – Jugular Venous Pulse Waveform.

Parameter	Values, P-876K015	Values, [19]	Values, [25]	Values, [26]
$\Delta x-a$ (ccf)	0.43 ± 0.02	0.30 ± 0.06	0.39 ± 0.09	0.33 ± 0.06
$\Delta y-v$ (ccf)	0.14 ± 0.02	0.12 ± 0.04	0.14 ± 0.03	0.11 ± 0.05
$\Delta c-a$ (ccf)	0.19 ± 0.02	-	-	-
$\Delta c-R$ (ccf)	0.11 ± 0.01	-	0.14 ± 0.04	-
$\Delta x-T$ (ccf)	0.09 ± 0.02	-	-	0.12 ± 0.06
$\Delta v-T$ (ccf)	0.20 ± 0.02	0.23 ± 0.05	0.23 ± 0.06	-
$\Delta y-T$ (ccf)	0.34 ± 0.02	0.35 ± 0.05	-	-

4 Discussion

This research builds upon previous investigations aimed at detecting cardiovascular parameters in the cervical region. Indeed, the common carotid arteries and the internal jugular veins are the main vessels for the blood flow in the heart-brain axis. Particularly, the JVP is an essential signal for the evaluation of the onset of neurodegenerative diseases. Nowadays, the JVP can be evaluated by using ultrasound and/or a blood pressure catheter, in hospital.

With the proposed study, we aim to preliminary evaluate the use of a piezoelectric thin film sensor as a sensing element sandwiched between two layers of adhesive silicone gel. Measurements were carried out to find the proper neck area for the application of the sensor. Between sternum, and jugular J1 and J2 segments, we chose J2 as the best. When the subject was seated, the pulse waveform detected was the arterial waveform and analysis of the data confirmed that the proposed method conformed to those found in the literature. The calculated parameters in Table 1 indicate good vascular health, with values such as VAI, MS Delay and P2-P1

Delay agreeing with those reported in the literature for subjects with similar age group. Although the b/a , c/a and e/a amplitude ratios differ numerically from those reported in studies based on optical PPG, they still follow the same relative trends and maintain physiologically significant ratios. These differences are most likely due to the use of a piezoelectric sensor that measures mechanical deformation of the vascular wall on the skin rather than changes in blood volume through light absorption. This method of detection can alter the amplitude profile of the signal derivatives, especially in the second derivative, while maintaining the temporal structure.

For the parameters listed in Table 2, our results are like those reported in literature for the $\Delta x-a$, $\Delta y-v$, $\Delta c-R$, $\Delta x-T$, $\Delta v-T$, $\Delta y-T$ time differences. Moreover, we provide the value also for the $\Delta c-a$ time difference, which was not possible to evaluate in [19,26] because of the low sampling rate of the ultrasound machine used to record the data.

5 Limitations and Future Work

This study focused on the detection of the blood pulse waveform at the neck level of a single male person. Although the detected waveform is as expected for both the arterial and venous signal, it is necessary to increase the number of subjects in the study to confirm these preliminary results. Moreover, comparing the signal acquired at different locations on the body can be useful to validate the detection of the arterial signal and give its result in mmHg. Most importantly, the results of this study may provide new insights into the current research field on venous pulse monitoring in a non-invasive and non-operator-dependent manner. In addition, our study focused on detecting the blood pulse waveform when the subject was supine and seated. To improve the research, the next study should include longer measurements and movement exercise.

6 Conclusion

Our study, aimed at testing the goodness of an epidermal patch for blood pulse detection at the neck level, suggests placing the piezoelectric thin film sensor at J2 segment of the right internal jugular vein. On this cervical area, the proposed solution was able to acquire the signals of interest, and the signal processing technique used for their data analysis confirms the results of the current literature. Utilizing a piezoelectric thin film sensor with an area larger than 1 cm^2 is recommended to enable independent use of the patch without requiring assistance from a healthcare operator. A smaller sensor may not be precisely positioned on the spot, leading to inaccurate pulse detection. Conversely, a larger sensor improves placement reliability, ensuring more consistent and accurate measurements for personal health applications.

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